

CS 280 Examination

Discrete Mathematics and Combinatorics

Instructions:

- Answer all questions.
 - For Questions 1–4, choose the best option.
 - For Questions 5–8, write brief but precise answers.
1. Which of the following best describes the relationship between a graph being bipartite and its chromatic number?
 - (A) A bipartite graph always has chromatic number 2
 - (B) A bipartite graph has chromatic number at most 2
 - (C) A graph is bipartite if and only if its chromatic number is even
 - (D) A graph with chromatic number 2 must contain a cycle
 2. In the context of generating functions, what does the coefficient of x^n in the expansion of $(1 - x)^{-k}$ represent?
 - (A) The number of ways to partition n into k parts
 - (B) The number of ways to choose n objects from k types with repetition
 - (C) $\binom{n+k-1}{k-1}$, the number of non-negative integer solutions to $x_1 + \cdots + x_k = n$
 - (D) The n -th Catalan number
 3. What is the negation of the statement $\forall x \exists y P(x, y)$?
 - (A) $\exists x \forall y \neg P(x, y)$
 - (B) $\forall x \forall y \neg P(x, y)$
 - (C) $\exists x \exists y \neg P(x, y)$
 - (D) $\neg \forall x \forall y P(x, y)$
 4. Which of the following is a valid application of the Pigeonhole Principle?
 - (A) If 13 people are in a room, at least two were born in the same month
 - (B) If n objects are placed in n boxes, each box contains exactly one object
 - (C) If a graph has more edges than vertices, it must contain a cycle

- (D) In any sequence of $n^2 + 1$ distinct real numbers, there exists a strictly increasing or strictly decreasing subsequence of length $n + 1$
5. Prove that a graph is bipartite if and only if it contains no odd-length cycles. Use this to explain why the complete graph K_n is bipartite only when $n \leq 2$.
 6. Explain the difference between a permutation and a combination. How does the inclusion-exclusion principle extend these basic counting techniques to more complex scenarios?
 7. State and prove the Handshaking Lemma for graphs. What immediate corollary follows regarding the number of vertices with odd degree?
 8. Describe the difference between strong induction and weak induction. Provide an example of a problem where strong induction is more natural to apply than weak induction.